

Practical 1.1.1

Algorithm:

Step 1 -> start

Step 2 -> take input of radius

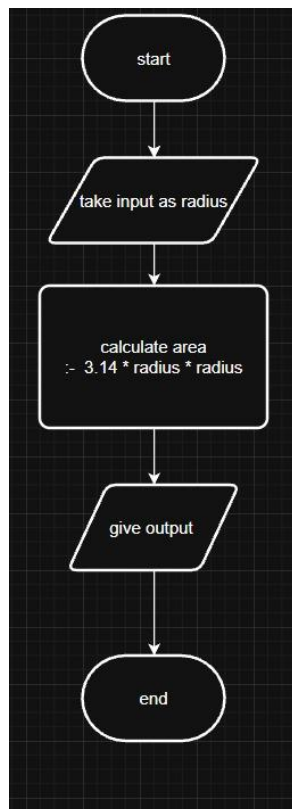
Step 3 -> calculate area of circle as:

$$\text{Area} = 3.14 * \text{radius} * \text{radius}$$

Step 4 -> give output

Step 5 -> stop

Flowchart:



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1.1.1. Area of Circle

06:58

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Write a Python program that calculates the area of a circle when the radius is provided by the user. Use $\pi = 3.14$ and display the area.

Input Format:

- A single line containing a floating-point number representing the radius.

Output Format:

- Print the computed area of the circle formatted to 4 decimal places.

Sample Test Cases

Test case 1

3.36

35.4493

Test case 2

2

12.5600

circlearea...

Submit

Explorer

Debugger

```
1 radius = float(input())
2
3 pi = 3.14
4
5 area = pi*radius*radius
6
7 print(f"area:.4f")
```

Average time

0.009 s

9.60 ms

Maximum time

0.013 s

13.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1

7 ms

Debug

⋮

⌵

Expected output

Actual output

3.36

3.36

35.4493

35.4493

Test case 2

8 ms

⌵

Terminal

Test cases

Practical 1.1.2

Algorithm:

Step 1 -> start

Step 2 -> take input of length and width

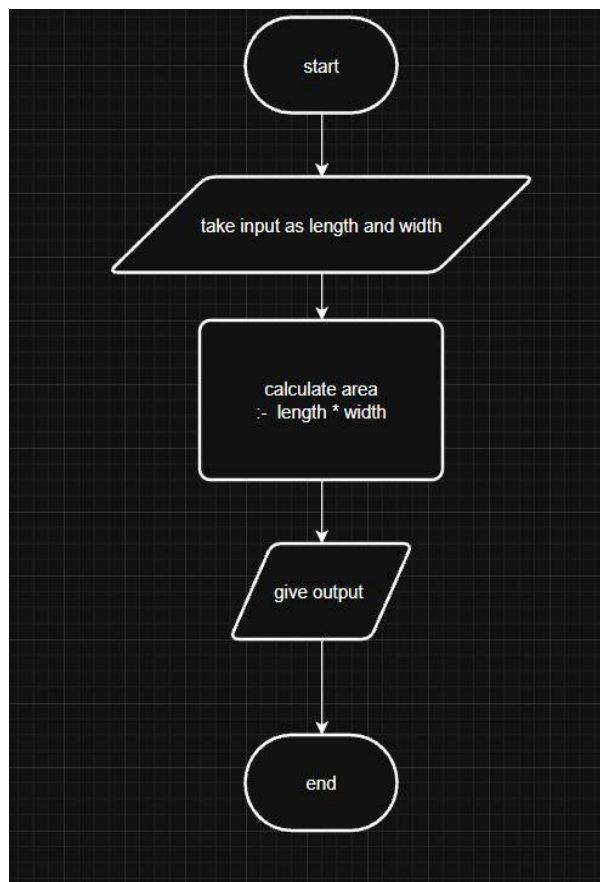
Step 3 -> calculate area :-

$$\text{Area} = \text{length} * \text{width}$$

Step 4 -> give output

Step 5 -> stop

Flowchart:



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1.1.2. Area of Rectangle

0123

Write a Python program to calculate the area of a rectangle given its length and width.

Formula:
Area of Rectangle = Length $\times$ Width

Input Format:

- First line contains a float value representing the length of the rectangle
- Second line contains a float value representing the width of the rectangle

Output Format:

- Print the area of the rectangle as a float value formatted to 2 decimal places.

Sample Test Cases

areaOfRe...

```
1
2 length = float(input())
3 width = float(input())
4
5
6 area = length * width
7
8
9 print(f"area:.2f")
10
11
```

Average time
0.011 s
11.20 ms

Maximum time
0.013 s
13.00 ms

5 out of 5 shown test case(s) passed
5 out of 5 hidden test case(s) passed

Test case 1 13 ms

Expected output
18.5
5.2
54.60

Actual output
18.5
5.2
54.60

Test case 2 0 ms

Terminal

Test cases

Practical 1.1.3

Algorithm:

Step 1 -> start

Step 2 -> take input of side_length

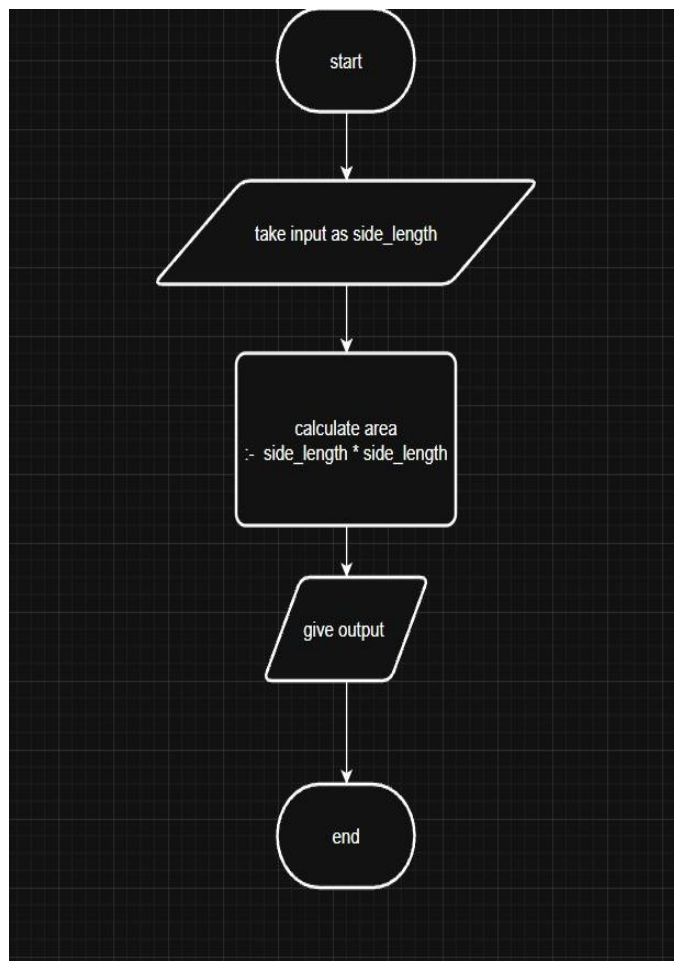
Step 3 -> calculate area :-

$$\text{Area} = \text{side_length} * \text{side_length}$$

Step 4 -> give output

Step 5 -> stop

Flowchart:



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1.1.3. Calculate Area of the Square

Write a Python program that prompts the user to enter the *side_length* of a square and computes the area of the square.

Formula:

- Area = side_length^2

Input Format:

- The input is a positive integer value that represents the *side_length* of the square.

Output Format:

- The output is a positive integer value that represents the area of the square.

Sample Test Cases

Test case 1

5

25

Test case 2

6

36

AreaSqua...

```
1
2 side = int(input())
3
4 area = side * side
5
6
7 print(area)
8
9
10
11
```

Average time

0.005 s

4.75 ms

Maximum time

0.006 s

6.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1

6 ms

Expected output

5

25

Actual output

5

25

Test case 2

6 ms

Terminal

Test cases

Practical 1.1.4

Algorithm:

Step 1 -> start

Step 2 -> take input of base and height

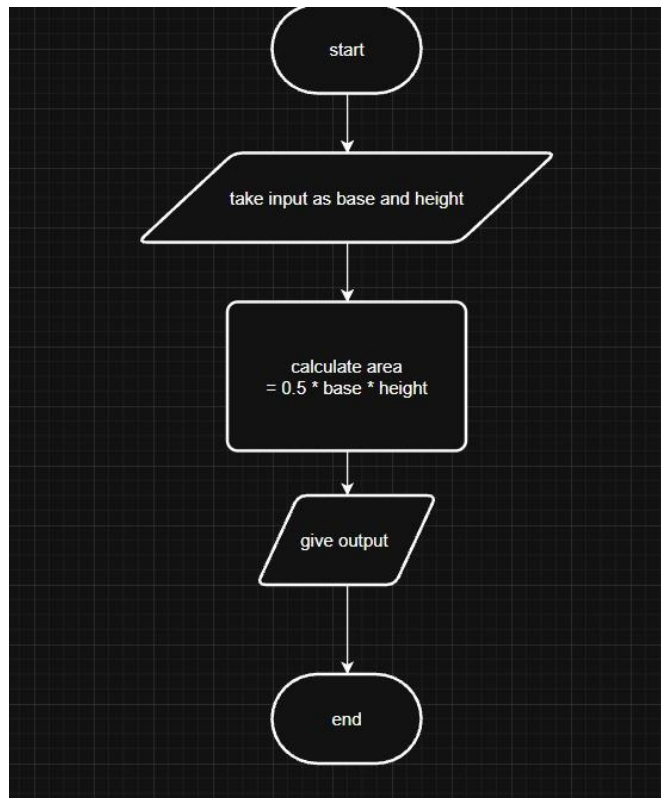
Step 3 -> calculate area :-

$$\text{Area} = 0.5 * \text{base} * \text{height}$$

Step 4 -> give output

Step 5 -> stop

Flowchart:



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1.1.4. Area of Triangle

00:29

Write a Python program that prompts the user to enter the triangle's base and height and computes the triangle's area.

Formula: $\text{Area of Triangle} = 0.5 \times \text{base} \times \text{height}$.

Input Format:

- The first line of input is the float value that represents the base of the triangle.
- The second line of input is the float value that represents the height of the triangle.

Output Format:

- The output is the floating point value that represents the area of a triangle, formatted to two decimals.

Sample Test Cases

Test case 1

6.54

1.23

4.02

Test case 2

10.5

32.0

168.00

triangleA...

Submit

```
1 base = float(input())
2
3 height = float(input())
4
5 area = 0.5*base*height
6
7 print(f"{area:.2f}")
```

Average time0.008 s8.00 ms

Maximum time0.009 s9.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 18 ms

Debug

Expected output

6.54

1.23

4.02

Actual output

6.54

1.23

4.02

Test case 27 ms

Terminal

Test cases

Practical 1.1.5

Algorithm:

Step 1 -> start

Step 2 -> take input of marks

Step 3 -> if marks ≥ 40

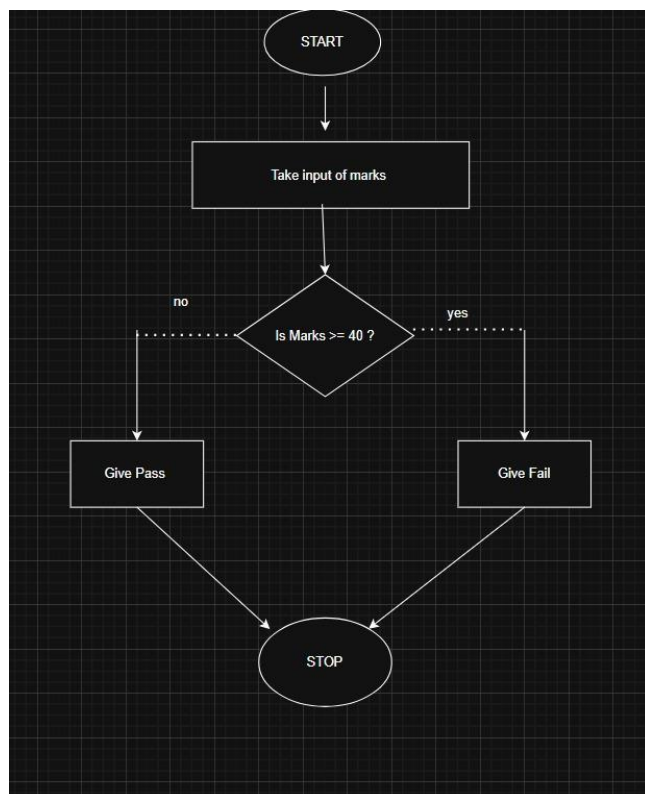
Step 4 -> give pass

Step 5 -> else

Step 6 -> give fail

Step 7 -> stop

Flowchart:



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1.1.5. Student Pass or Fail Status

01:35

Write a Python program to determine whether a student passed the exam or not based on their marks.

Pass/Fail Criteria:

A student passes if marks ≥ 40

A student fails if marks < 40

Input Format:

Single line contains an integer representing the marks obtained by the student.

Output Format:

Print "Pass" if the student passed the exam.

Sample Test Cases

Test case 1

45

Pass

Test case 2

35

Fail

Test case 3

40

Pass

passOrFa...

1 marks = int(input())

2

3 if marks >= 40:

4 print("Pass")

5 else:

6 print("Fail")

Average time

0.005 s

5.29 ms

Maximum time

0.008 s

8.00 ms

3 out of 3 shown test case(s) passed

4 out of 4 hidden test case(s) passed

Test case 1 8 ms

Expected output

45

Actual output

45

Pass

Test case 2 4 ms

Test case 3 4 ms

Terminal

Test cases

< Prev

Reset

Submit

Next >

Practical 2.1.1

Algorithm:

Step 1 → Start

Step 2 → Input coefficients a, b, and c

Step 3 → Calculate the discriminant

$$D = b^2 - 4ac$$

Step 4 → If $D > 0$

→ Calculate

$$\text{root1} = (-b + \sqrt{D}) / (2a)$$

$$\text{root2} = (-b - \sqrt{D}) / (2a)$$

→ Print roots as real and different (to two decimal places)

Step 5 → Else if $D == 0$

→ Calculate

$$\text{root} = -b / (2a)$$

→ Print roots as real and same (to two decimal places)

Step 6 → Else ($D < 0$)

→ Calculate

$$\text{realPart} = -b / (2a)$$

$$\text{imaginaryPart} = \sqrt{-D} / (2a)$$

→ Print roots as imaginary

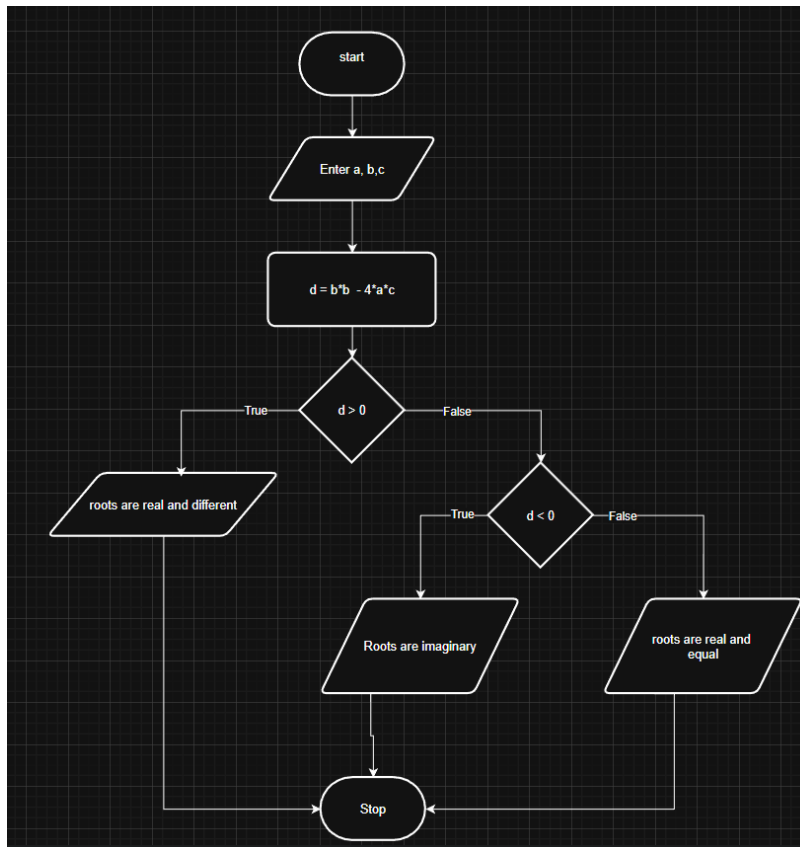
$$\text{root1} = \text{realPart} + \text{imaginaryPart } i$$

$$\text{root2} = \text{realPart} - \text{imaginaryPart } i$$

(formatted to two decimal places)

Step 7 → Stop

Flowchart:



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2.1.1. Roots of a Quadratic Equation

Write a program to find the roots of a quadratic equation, given its coefficients a , b , and c . Use the quadratic formula: $\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

The discriminant $D = b^2 - 4ac$ determines the nature of the roots:

- If $D > 0$: Roots are real and different
- If $D = 0$: Roots are real and the same
- If $D < 0$: Roots are imaginary

Sample Test Cases

Test case 1

1 -5 6

root1 = 3.00

root2 = -2.00

Test case 2

1 -4 4

root1 = root2 = 2.00

Test case 3

1 2 5

root1 = -1.00+2.00i

root2 = -1.00-2.00i

quadratic...

```

1 import math
2
3 # Read coefficients a, b, c
4 a, b, c = map(int, input().split())
5
6 # Calculate discriminant
7 d = b * b - 4 * a * c
8
9 if d > 0:
10     root1 = (-b + math.sqrt(d)) / (2 * a)
11     root2 = (-b - math.sqrt(d)) / (2 * a)
12     print(f"root1 = {root1:.2f}")
13     print(f"root2 = {root2:.2f}")
14
15 elif d == 0:
16     root = -b / (2 * a)
17     print(f"root1 = root2 = {root:.2f}")
  
```

Average time: 0.010 s
Maximum time: 0.011 s
9.83 ms 11.00 ms

3 out of 3 shown test case(s) passed
3 out of 3 hidden test case(s) passed

Test case 1 11 ms

Expected output: 1 -5 6

Actual output: root1 = 3.00, root2 = -2.00

Test case 2 8 ms

[Terminal](#)
[Test cases](#)

Practical 3.1.1

Algorithm:

Step 1 → Start

Step 2 → Input three integers a, b, and c

Step 3 → If $a \geq b$ **and** $a \geq c$

→ Print a as the largest number

Step 4 → Else if $b \geq a$ **and** $b \geq c$

→ Print b as the largest number

Step 5 → Else

→ Print c as the largest number

Step 6 → Stop

Flowchart:

Algorithm:

Step 1 → Start

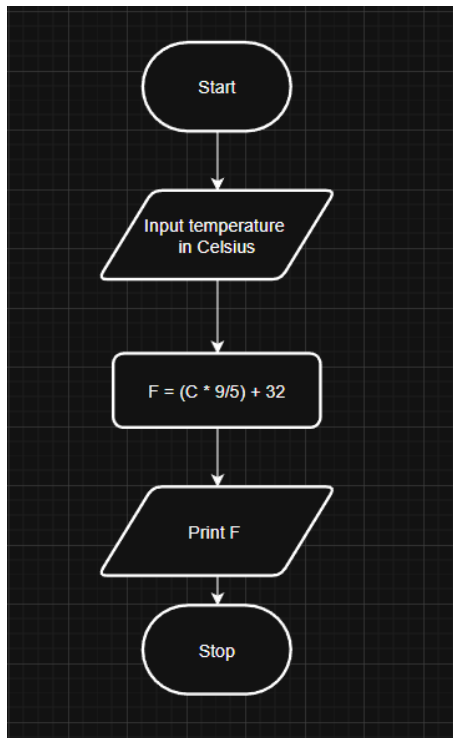
Step 2 → Input temperature in Celsius C

Step 3 → Calculate temperature in Fahrenheit using the formula
 $F = (C \times 9/5) + 32$

Step 4 → Print F formatted to two decimal places

Step 5 → Stop

Flowchart:



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3.1.2. Celsius to Fahrenheit

Write a Python program to convert temperature from Celsius to Fahrenheit.

Formula:

$$\text{Fahrenheit} = \left(\text{Celsius} \times \frac{9}{5} \right) + 32$$

Input Format:
 • Single line contains a float value representing the temperature in Celsius.

Sample Test Cases

Test case	Input	Output
Test case 1	0.0	32.00
Test case 2	37.5	99.50
Test case 3	-40	-40.00
Test case 4	25	77.00

```

1 celsius = float(input())
2
3 fahrenheit = (celsius * 9 / 5) + 32
4
5 print(f"fahrenheit: .2f")
6
7
  
```

Performance:
 Average time: 0.009 s (9.50 ms) | Maximum time: 0.012 s (12.00 ms)
 4 out of 4 shown test case(s) passed
 4 out of 4 hidden test case(s) passed

Test case	Expected output	Actual output
Test case 1 (9 ms)	0.0 32.00	0.0 32.00
Test case 2 (7 ms)		
Test case 3 (11 ms)		

Terminal Test cases

Practical 4.1.1

Algorithm:

Step 1 → Start

Step 2 → Input elements of **Set A** as space-separated integers

Step 3 → Input elements of **Set B** as space-separated integers

Step 4 → Convert the input lists into sets A and B

Step 5 → Find the **union** of the sets

→ Union = $A \cup B$

Step 6 → Find the **intersection** of the sets

→ Intersection = $A \cap B$

Step 7 → Find the **difference** of the sets

→ Difference = $A - B$

Step 8 → Print

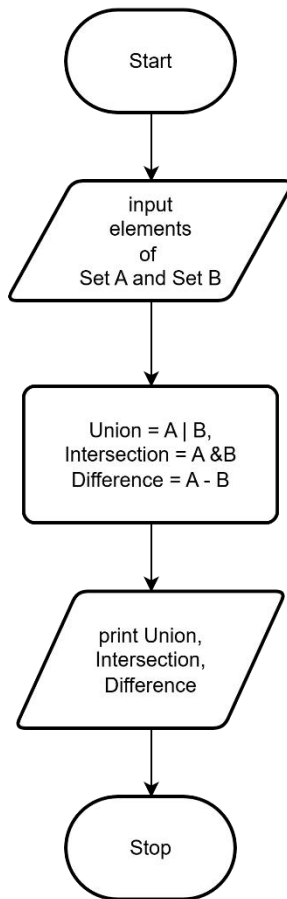
→ "Union: " followed by the union set

→ "Intersection: " followed by the intersection set

→ "Difference: " followed by the difference set

Step 9 → Stop

Flowchart:



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4.1.1. Set Operations

Write a Python program to perform union, intersection and difference operations on *Set A* and *Set B*.

Input Format:

- First Line prompts "Set A: " followed by space-separated list of integers for *Set A*.
- The second input prompts "Set B: " followed by space-separated list of integers for *Set B*.

Output Format:

- The first line prints "Union: " followed by the union of *Set A* and *Set B*.

Sample Test Cases

Test case 1

Set A: 0 2 4 5 8

Set B: 1 2 3 4 5

Union: {0, 1, 2, 3, 4, 5, 8}

Intersection: {2, 4, 5}

Difference: {0, 8}

Test case 2

Set A: 10 11 22 33 44 55

Set B: 15 16 17 18 19 14

Union: {33, 10, 11, 44, 14, 15, 16, 17, 18, 19, 22, 55}

Intersection: set()

Difference: {33, 10, 11, 44, 22, 55}

```

1 # Read input with prompts
2 set_a = set(map(int, input("Set A: ").split()))
3 set_b = set(map(int, input("Set B: ").split()))
4
5 # Perform set operations
6 union_set = set_a | set_b
7 intersection_set = set_a & set_b
8 difference_set = set_a - set_b
9
10 # Print results
11 print("Union:", union_set)
12 print("Intersection:", intersection_set)
13 print("Difference:", difference_set)
14
15
  
```

Average time: 0.012 s
Maximum time: 0.014 s
11.75 ms 14.00 ms

2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 1 14 ms

Expected output	Actual output
Set A: 0 2 4 5 8	Set A: 0 2 4 5 8
Set B: 1 2 3 4 5	Set B: 1 2 3 4 5
Union: {0, 1, 2, 3, 4, 5, 8}	Union: {0, 1, 2, 3, 4, 5, 8}
Intersection: {2, 4, 5}	Intersection: {2, 4, 5}
Difference: {0, 8}	Difference: {0, 8}

Terminal Test Cases

Practical 5.1.1

ALGORITHM:

Step 1 : Start

Step 2 : Input year y

Step 3 : Check if $y \% 4 == 0$

Step 4 : If yes

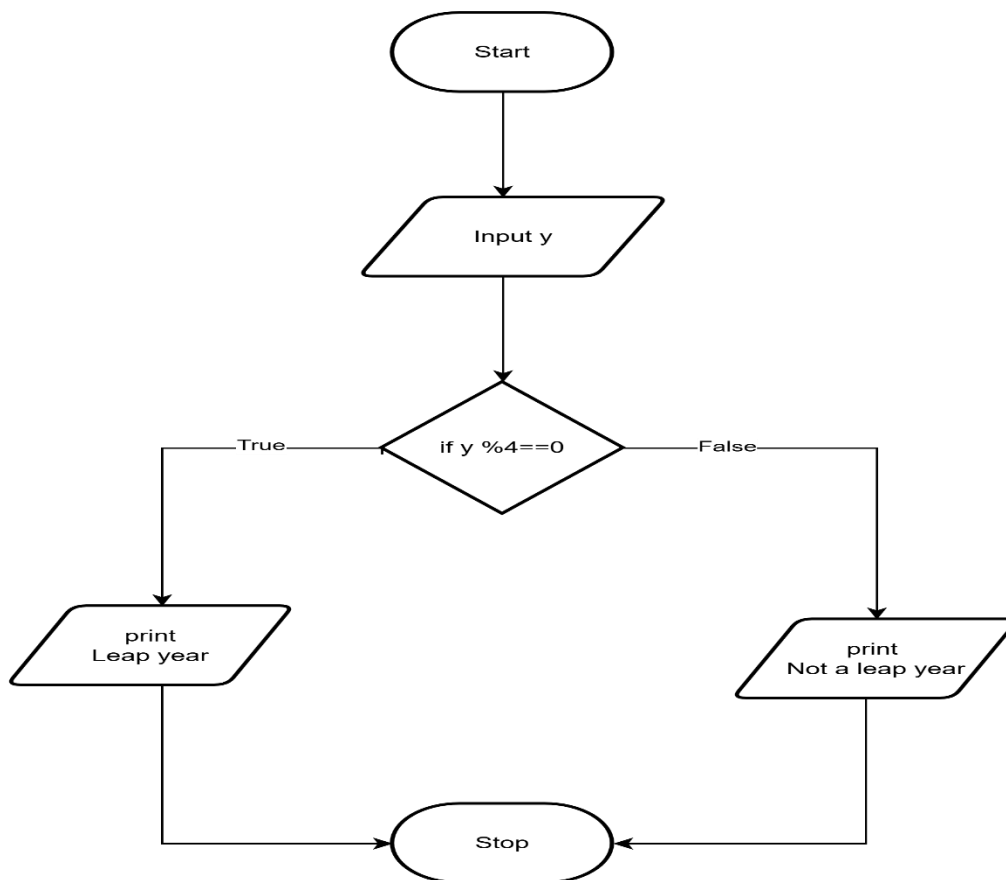
Print "Leap year"

Step 5 : Else

Print "Not a leap year"

Step 6 : Stop

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5.1.1. Leap Year Checker

02/05

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Write a Python program that prompts the user to enter a year. The program should determine if the year is a leap year or not and print the appropriate message.

Input Format:

- A single line contains an integer representing the year.

Output Format:

- Print "Leap year" if it is a leap year. Otherwise, print "Not a leap year".

Sample Test Cases

Test case 1

2024

Leap year

Test case 2

2022

Not a leap year

leapYear.py

Submit

Debugger

```
1 a = int(input())
2 if a % 4 == 0:
3     print("Leap year")
4 else:
5     print("Not a leap year")
```

Average time

0.009 s

9.00 ms

Maximum time

0.011 s

11.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1

9 ms

Debug

⌵

Expected output

2024

Leap year

Actual output

2024

Leap year

Test case 2

7 ms

⌵

Terminal

Test cases

Practical 5.1.2

ALGORITHM

Step 1: Start

Step 2: Input four marks: a, b, c, d

Step 3: Compute total

$$\text{total} = a + b + c + d$$

Step 4: Compute percentage

$$\text{percent} = (\text{total} / 400) \times 100$$

Step 5: Print total

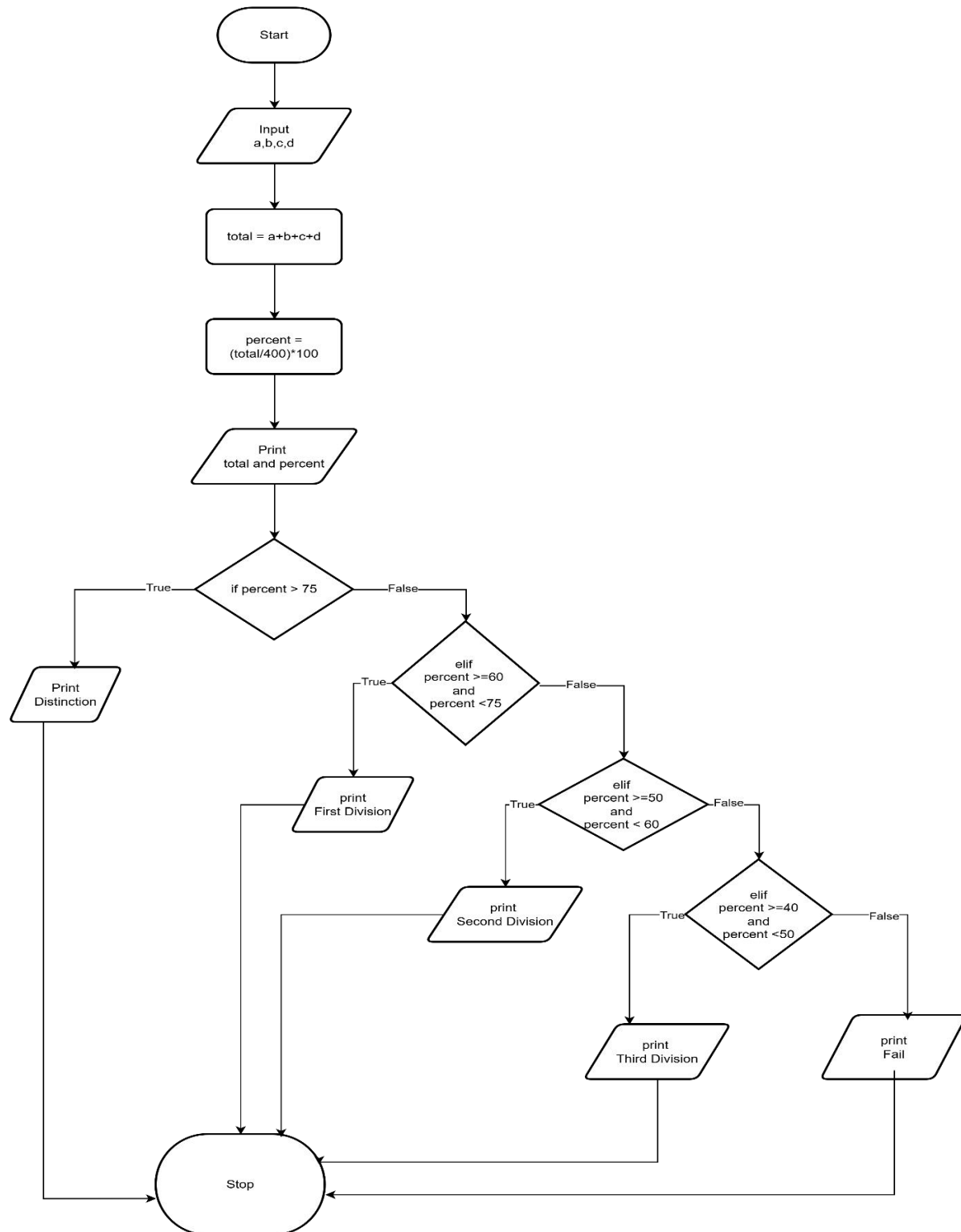
Step 6: Print percent (up to 2 decimal places)

Step 7: Check conditions:

- If percent > 75
Print Distinction
- Else if percent $\geq$ 60
Print First Division
- Else if percent $\geq$ 50
Print Second Division
- Else if percent $\geq$ 40
Print Third Division
- Else
Print Fail

Step 8: Stop

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5.1.2. Student Grade Based on Aggregate18.5%

Sample Test Cases

Test case 1

85 90 78 88

341

85.25

Distinction

Test case 2

65 70 60 68

263

65.75

First Division

Test case 3

55 50 52 58

215

53.75

Second Division

Test case 4

45 42 40 48

studentG...

```
1 a,b,c,d = map(int,input().split())
2 total = a+b+c+d
3 percent = float((total/400)*100)
4 print(total)
5 print(f'{percent:.2f}')
6 if percent > 75:
7     print("Distinction")
8 elif percent >= 60 and percent < 75:
9     print("First Division")
10 elif percent >= 50 and percent < 60:
11     print("Second Division")
12 elif percent >= 40 and percent < 60:
13     print("Third Division")
14 else:
15     print("Fail")
```

Average time0.012 s11.80 ms

Maximum time0.022 s22.00 ms

5 out of 5 shown test case(s) passed

5 out of 5 hidden test case(s) passed

Test case 1412 ms

Expected output

Actual output

85 90 78 88

341

85.25

Distinction

Test case 241 ms

Terminal

Test cases

Practical 6.1.1

ALGORITHM

Step 1 Start

Step 2 Input d, m, y

Step 3 Set flag $\leftarrow 1$

Step 4 If $y < 0$ OR $m < 1$ OR $m > 12$, set flag $\leftarrow 0$

Step 5 Determine maxdays:

- If m in $\{1,3,5,7,8,10,12\} \rightarrow 31$
- If m in $\{4,6,9,11\} \rightarrow 30$
- Else (February):
 - If $y \% 2 == 0 \rightarrow 29$
 - Else $\rightarrow 28$

Step 6 If $d < 1$ OR $d > \text{maxdays}$, set flag $\leftarrow 0$

Step 7 If flag == 1:

- $d \leftarrow d + 1$
- If $d > \text{maxdays}$:
 - $d \leftarrow 1, m \leftarrow m + 1$
- If $m > 12$:
 - $m \leftarrow 1, y \leftarrow y + 1$
- Print date in dd-mm-yyyy format

Step 8 Else print "Invalid Date"

Step 9 End

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6.1.1. Incremented Date

Write a Python program to check if a given date is valid. If the date is valid, print the next day's date (incremented date). If the date is invalid, print "Invalid Date".

Date Validation Rules:

- Valid Month:** 1 to 12
- Valid Day:** Depends on the month and year
 - January (1), March (3), May (5), July (7), August (8), October (10), December (12): 1 to 31 days
 - April (4), June (6), September (9), November (11): 1 to 30 days
 - February (2): 1 to 29 days in a leap year and 1 to 28 days in a non-leap year
- Valid Year:** Any positive integer greater than zero.

Date Increment Rules:

If the date is valid, increment it by one day:

- If it's the last day of the month, move to the 1st day of the next month
- If it's December 31st, move to January 1st of the next year

Input Format:

- First line contains an integer representing the day
- Second line contains an integer representing the month
- Third line contains an integer representing the year

Output Format:

- If the date is valid, print the incremented date in the format: `<DD>-<MM>-<YYYY>`
- If the date is invalid, print: "Invalid Date".

Sample Test Cases

```

nextDate.py
1 d=int(input())
2 m=int(input())
3 y=int(input())
4
5 flag = 1
6 if y<0 or m<0 or m>12:
7     flag = 0
8 if m in (1,3,5,7,8,10,12):
9     maxdays = 31
10 elif m in (4,6,9,11):
11     maxdays = 30
12 else:
13     if y % 4 == 0:
14         maxdays = 29
15     else:
16         maxdays = 28
17

```

Average time
0.020 s
 20.10 ms

Maximum time
0.063 s
 63.00 ms

5 out of 5 shown test case(s) passed
 5 out of 5 hidden test case(s) passed

Test case 1 63 ms
 [Debug](#)

Expected output	Actual output
15	15
3	3
2024	2024
16-03-2024	16-03-2024

Test case 2 63 ms

[Terminal](#)
[Test cases](#)

Practical 6.1.2

ALGORITHM:

Step 1: Input integer a

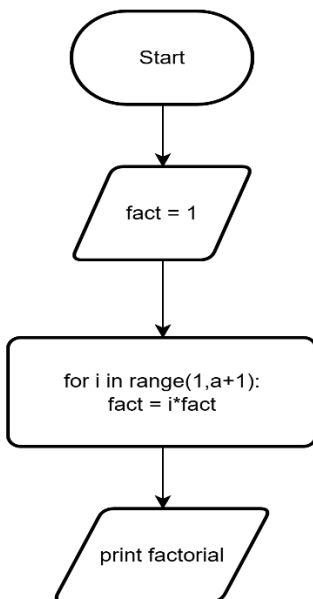
Step 2: □ Set $\text{fact} \leftarrow 1$

Step 3: For i from 1 to a do:
 $\text{fact} \leftarrow \text{fact} \times i$

Step 4: Print fact

Step 5: End

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6.1.2. Factorial of a Number

11/43

Write a Python program to calculate the factorial of a number n using loops.

Input Format:

A single line containing an integer n .

Output Format:

Print the factorial of the given integer n .

Sample Test Cases

Test case 1

10

3628800

Test case 2

3

6

factorialN...

Submit

1a = int(input())

2fact = 1

3for i in range(1,a+1):

4fact = fact*i

5

6print(fact)

Average time0.005 s5.25 ms

Maximum time0.006 s6.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 16 ms

Debug

Expected output

10

3628800

Actual output

10

3628800

Test case 24 ms

Terminal

Test cases